



American International University-Bangladesh (AIUB)

Department of Computer Science Faculty of Science & Technology (FST)

Emergency Service System-Rapid Aid

A Software Engineering Project Submitted
By

Semester: Spring_24_25		Section: L	Group Number: 08	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	Abrar Rejwan Prince	22-48757-3		
2	Md Samiul Rayan Rifat	22-49975-3		
3	Faiyaz Azmaine	22-48804-3		
4	MD. Ashfak Uzzaman Khan Limon	22-49976-3		
5	Most. Sumaia Farzana Nitu	22-48243-2		

The project will be evaluated for the following Course Outcomes

CO3: <i>Select appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects</i>	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]	
Evidence of Argumentation Regarding Process Model Selection	[5Marks]	
Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	

CO4: Develop a project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	
	Total Marks	
CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain		
Taking project responsibility: perform assigned tasks on time independently	[5 Marks]	
Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	
Showing respect and value towards other team member's opinion	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: Abrar Rejwan Prince

Student ID: 22-48757-3

Contribution in Percentage (20%):

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

Signature of the Student

Student Name: Md Samiul Rayan Rifat

Student ID: 22-49975-3

Contribution in Percentage (20%):

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

Signature of the Student

Student Name: Faiyaz Azmaine

Student ID: 22-48804-3

Contribution in Percentage (20%):

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

Signature of the Student

Student Name: MD. Ashfak Uzzaman Khan Limon

Student ID: 22-49976-3

Contribution in Percentage (20%):

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

Signature of the Student
Student Name: Most. Sumaia Farzana Nitu Student ID: 22-48243-2 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 Signature of the Student

Section 1: UI/UX

In the early stages of application development, careful planning and design are crucial to ensuring the final product meets user expectations. To illustrate this process, the following image showcases a user interface design for our mobile application, **Rapid Aid**, created in Figma. This design sequence includes essential screens for user registration, login, SOS feature, and messaging functionalities, demonstrating how we establish a clear workflow from the user's first interaction with the app to more advanced features.

The series of screens visualized here follows the progression from initial user registration to the core application interface, where users can send SOS messages and communicate within a community. It highlights the importance of a clean, intuitive layout and interaction flow. Each screen serves as a key element in the user's journey.

For Simulation Experience on (Android/iOS): [Rapid Aid UI/UX simulation](#)

Layout and Design: The interface will prioritize simplicity and ease of navigation, allowing users to log in, browse, and access critical information effortlessly. Essential features such as SOS alerts, emergency calls (police, ambulance, doctor), and chatbots will be prominently accessible.

Usability: To ensure quick adoption, the design will incorporate intuitive icons, clear labels, and a visually distinct color scheme to guide users. Frequently used actions will be optimized with shortcuts and pre-configured settings for faster response times.



Figure 1: Home Screen App Icon

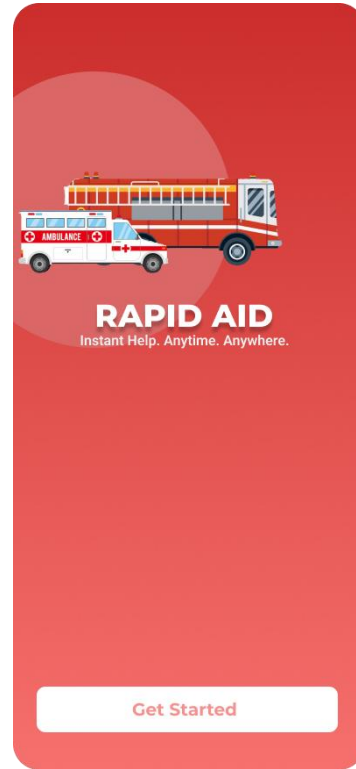


Figure 2: Starting window

Figure 3: Registration Page

Figure 4: Log in Page

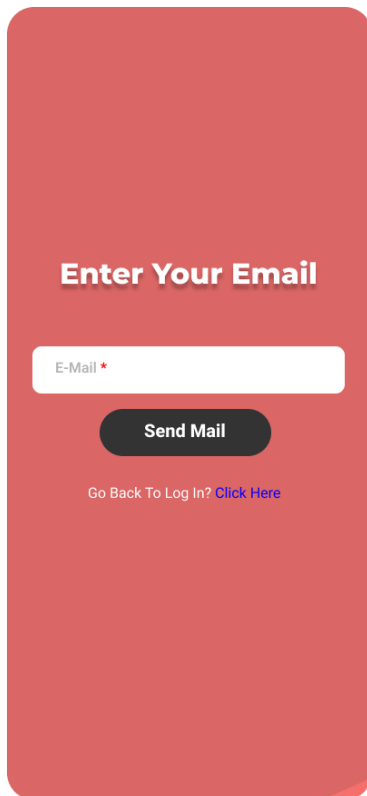


Figure 5: password reset email

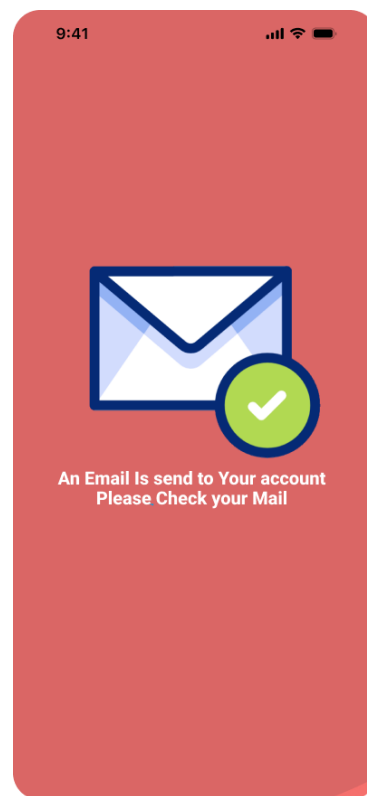


Figure 6: Email sending Confirmation

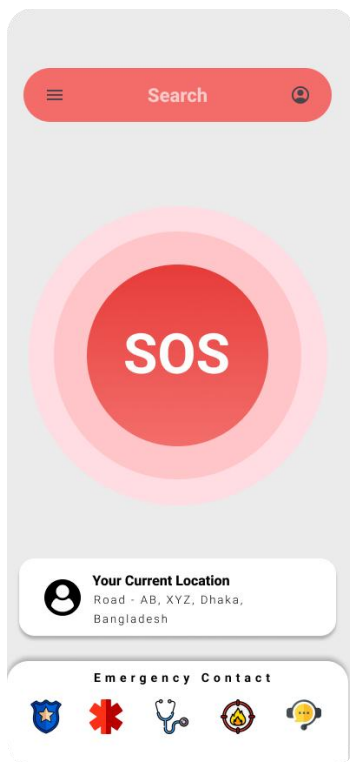


Figure 7: homepage

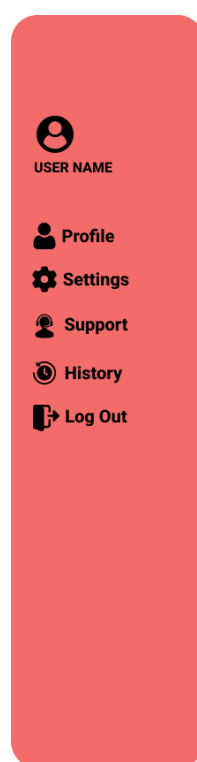


Figure 8: Sidebar

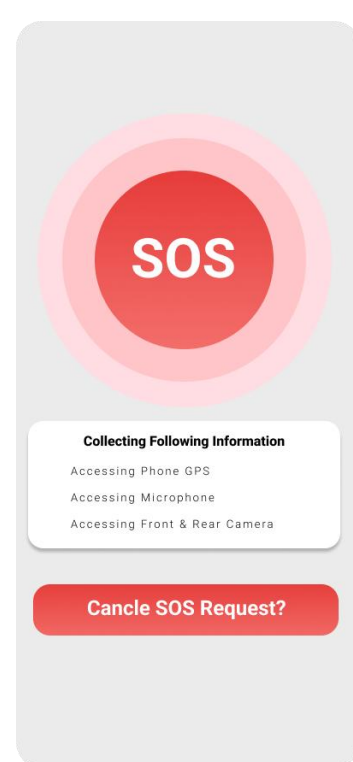


Figure 9: Emergency SOS



Figure 10: Police station Information

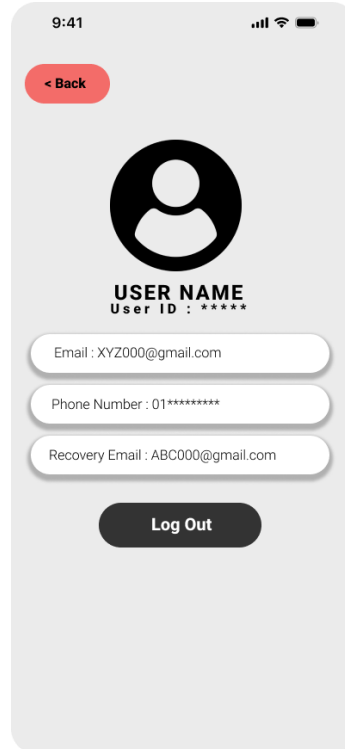


Figure 11: Profile

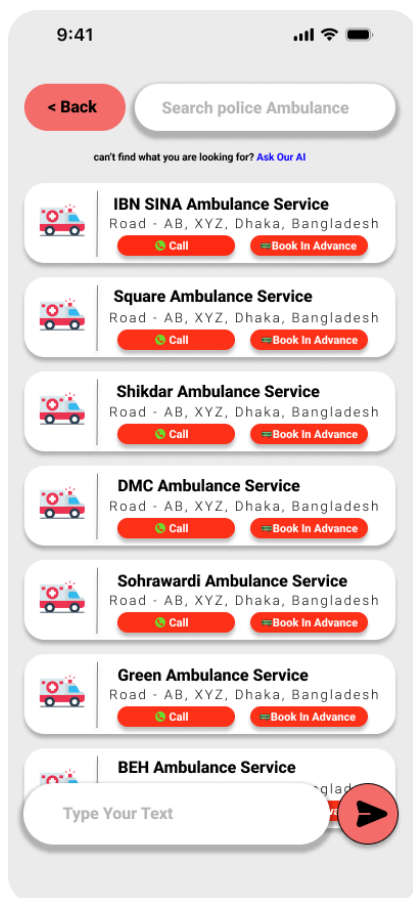


Figure 12: Ambulance Information

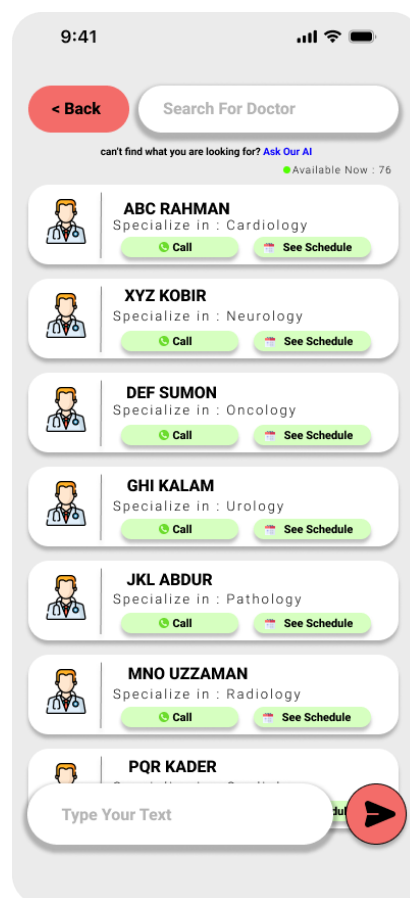


Figure 13: Doctor Information

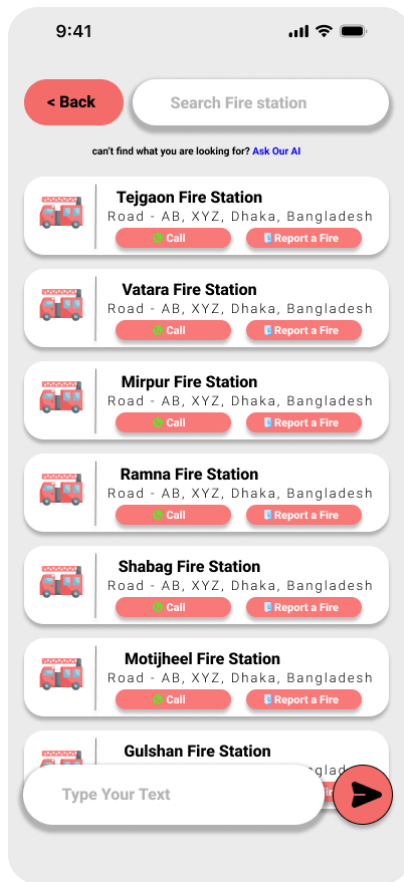


Figure 14: Fire station Information

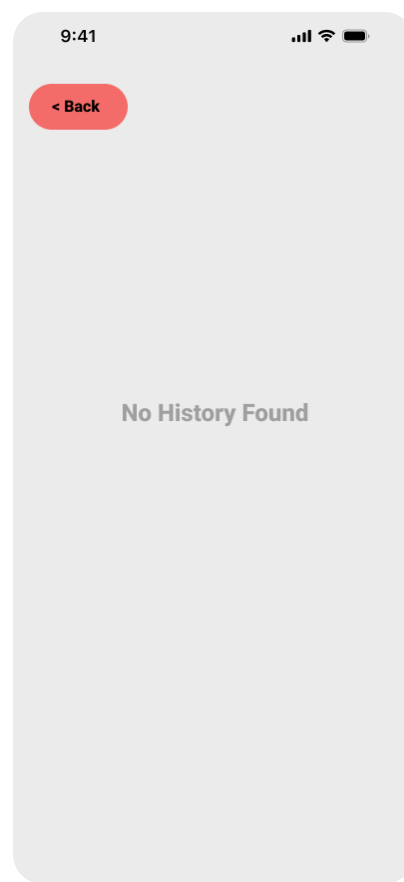


Figure 15: History of User

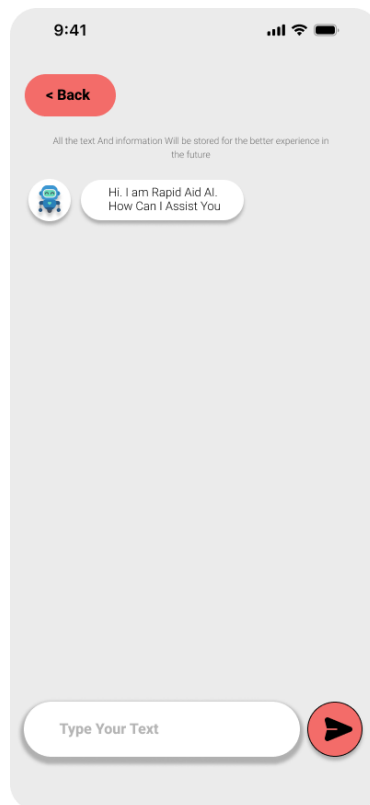


Figure 16: Rapid Aid AI Assistant

Section 2: Project Testing and Planning

2.1 Test Plan

We have selected functional requirements for testing in order to determine how the system should be tested. To do this, we have chosen the following requirements:

- 1.1 Unregistered username or password – Login should fail.
- 1.2 Valid user credentials – Login should be successful.
- 1.3 Invalid username or password – Login should fail.
- 1.4 Empty fields – An error message should be displayed.
- 1.5 Health Care Session-user should make a call with a healthcare consultant.
- 1.6 SOS signal: Used to get all nearby Emergency services
- 1.7 Verifying that the AI chatbot is working properly

2.2 Test Case

Test Case 1.1: Verify login fails for unregistered user

Project Name: Emergency Service System-Rapid Aid		Test Designed by: Lemon		
Test Case ID: 1.1		Test Designed date: 18/05/2025		
Test Priority ((Low, Medium, High): High		Test Executed by: Rifat		
Module Name: Unregistered username or password		Test Execution date: 18/05/2025		
Test Title	Verify login fails for unregistered user			
Description:	Test login functionality with an unregistered username.			
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1.Navigate to the login page. 2.Enter an unregistered username. 3.Click on the login button.	Username: unknown User Password: any Password	- Login page is displayed. - Login fails and 'User not found' error is displayed.	As expected,	pass
Post Condition: User remains on the login page with error displayed.				

Test Case 1.2: Verify login with valid username and password

Project Name: Emergency Service System-Rapid Aid		Test Designed by: Abrar		
Test Case ID:1.2		Test Designed date: 18/05/2025		
Test Priority (Low, Medium, High): High		Test Executed by: Rifat		
Module Name: Valid user credentials		Test Execution date: 18/05/2025		
Test Title	Verify login with valid username and password			
Description:	Test login functionality with valid credentials.			
Precondition (If any):	User has a valid registered account.			
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/ Fail)
1. Navigate to the login page. 2. Enter valid username. 3. Enter valid password. 4. Click on the login button.	Username: Rifat Password: 12345	- Login page is displayed. - User is logged in and redirected to the dashboard.	Login successfull.User got into the dashboard	Pass
Post Condition: User is validated and successfully logged into the application.				

Test Case 1.3: Verify login fails with invalid Data

Project Name: Emergency Service System-Rapid Aid	Test Designed by: Fayaiz
Test Case ID: 1.3	Test Designed date: 18/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Rifat

Module Name: Invalid username or password		Test Execution date: 18/05/2025		
Test Title	Verify login fails with invalid username or password			
Description:	Test login functionality with invalid username or password.			
Precondition (If any):	User account should not exists in the system.			
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the login page. 2. Enter invalid username. 3. Enter the invalid password. 4. . Click on the login button.	Username: valid Password: wrong Pass	Login fails and error messages are displayed.	Login fails and error message shown	Pass
Post Condition: User remains on the login page with error displayed.				

Test Case 1.4: Verify error message when login field empty

Project Name: Emergency Service System-Rapid Aid		Test Designed by: Fayaiz		
Test Case ID: 1.4		Test Designed date: 18/05/2025		
Test Priority ((Low, Medium, High): Medium		Test Executed by: Rifat		
Module Name: Empty fields		Test Execution date: 18/05/2025		
Test Title	Verify error messages when login fields are empty			
Description:	Test login functionality when both username and password fields are empty.			
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Navigate to the login page. 2.Leave username and password fields empty. 3. Click on the login button.	Username: blank Password: blank	Error message prompting to enter username and password is displayed.	A message box is shown asking for username and pass	pass
Post Condition: No login is performed; users are prompted to enter the required details.				

Test Case 1.5: Calling Health Care consultant

Project Name: Emergency Service System-Rapid Aid			Test Designed by: Rifat	
Test Case ID: 1.5 Test Priority (Low,Medium,High): High			Test Designed date: 22/05/2025 Test Executed by: Rifat	
Module Name: Health Care Session			Test Execution date: 22/05/2025	
Test Title: Request emergency consultant				
Description: This test case ensures that a user can request an emergency consultant by choosing a listed specialist and connecting with the specialist via video conference.				
Precondition (If any): 1. The user has a valid internet connection. 2. The user is logged into the system. 3. Location services are enabled.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the system 2. Select Heath Care services 3.Select Emergency Consultant 4.Choose one from the list of specialists 5.Confirm your request	Specialist: Cardiologist	The user successfully logs in, accesses health care services, selects an emergency consultant, chooses a specialist, and connects with them via call.	The user successfully logged in, accessed health care services, and selected a specialist and made a call .	Pass
Post Condition: 1. The information of the request is stored in the database. 2. A call is conducted between the user and the selected specialist.				

Test Case 1.6: Using SOS Signal for nearby Emergency

Project Name: Emergency services System-Rapid Aid	Test Designed by: Lemon
Test Case ID: 1.6	Test Designed date: 22-05-25
Test Priority (Low, Medium, High): High	Test Executed by: Rifat

Module Name: SOS signal implementation		Test Execution date: 22-05-25		
Test Title: show emergency services according to location				
Description: after giving SOS signal, user will see his nearby emergency services				
Precondition (If any):1. Valid internet connection 2.User Logged into system 3.Location service is enabled				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Login to the system 2.select SOS 3.Assign unit according to SOS	Request: SOS for nearby emergency Cause: Life Threating situation	User should get the proper location of nearby all emergency	After SOS selection, nearby emergency appered	PASS
Post Condition: Data base storation User can call related nearby emergency				

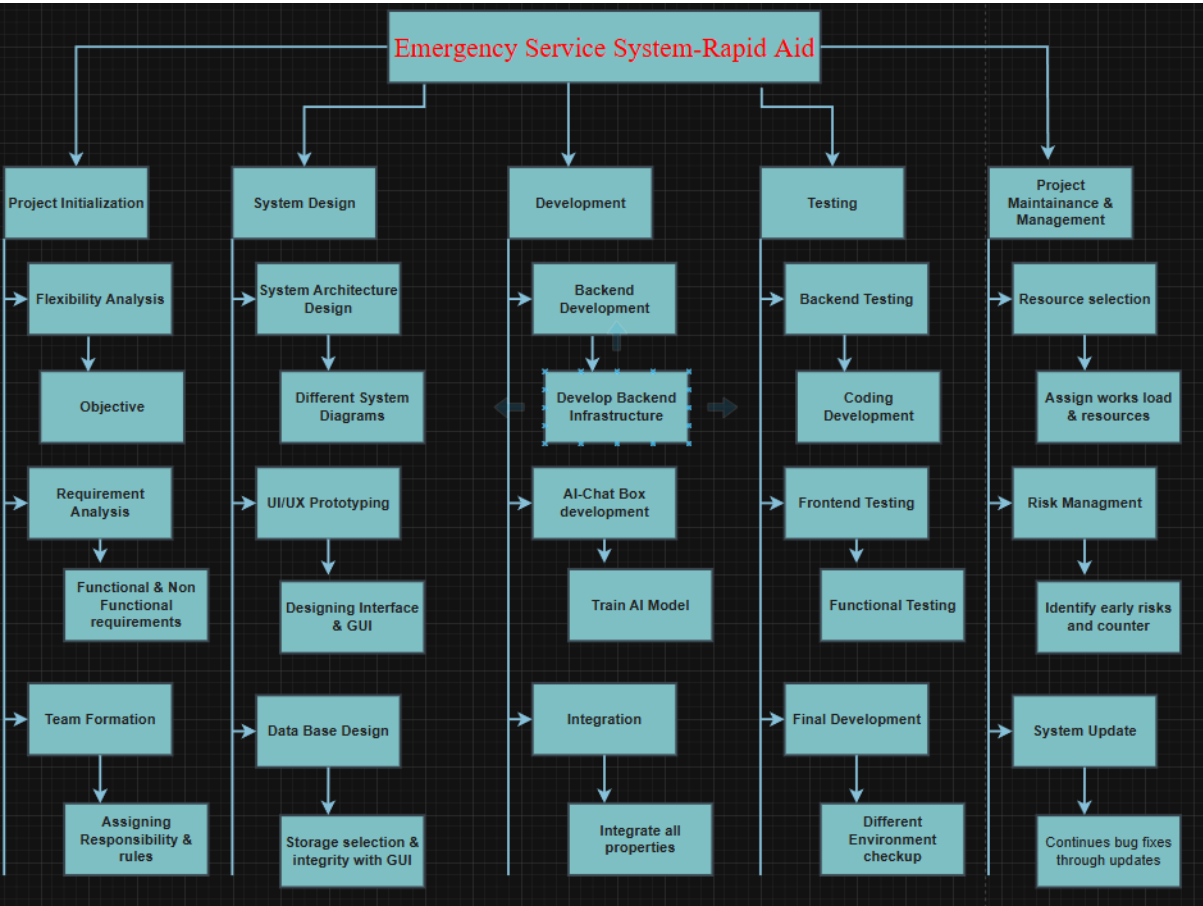
Test Case 1.7: Verify AI chatbot efficiency

Project Name: Emergency services System-Rapid Aid		Test Designed by: Lemon		
Test Case ID: 1.7		Test Designed date: 22-05-25		
Test Priority((Low, Medium, High): Low		Test Executed by: Rifat		
Module Name: Verify AI chatbot efficiency		Test Execution date: 22-05-25		
Test Title	Verify AI assigns emergency tasks based on user selection			
Description:	Test that AI assigns emergency based on user selection			
Precondition (If any):	User need to login to DashBoard			
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1.Login as a project manager 2.Navigate to the AI for emergency 3.Trigger the AI algorithm.	Username: valid, Password: 123 Task: i want a ambulance	User should get their desire emergency service details	Given call to nearby ambulance	Pass
Post Condition: Tasks are correctly assigned in the system according to the AI algorithm.				

Section 3: Project Schedule & Effort Estimation

3.1 WRO Model



3.2 Effort Estimation Using COCOMO Model

To estimate the effort and timeline for our Emergency Service System – Rapid Aid, we applied the COCOMO (Constructive Cost Model) using the Semi-Detached mode, as the project involves intermediate complexity and a team with mixed levels of experience.

Estimated LOC: Let's use SLOC = 11,000

B (Special Skills Factor) B: 1.8

P (Productivity Parameter) P: 800 SLOC/person-month

t (Project Duration) t = 4 months

$$\begin{aligned}\text{Effort Estimation, } E &= [\text{LOC} \times B^{0.333}/P]^3 \times (1/t^4) \\ &= [11000 \times 1.8^{0.333}/800]^3 \times (1/4^4) \\ &= 18.56 \text{ Person-months}\end{aligned}$$

Estimated total effort $\approx$ 19 person-months.

For a team of 5 people, this would take: **around 4 months** to develop the project.

$$\text{Effort (PM)} = \text{Coefficient (Effort Factor)} \times (\text{SLOC}/1000)^P$$

$$\text{Development Time (DM)} = 2.50 \times (\text{PM})^T$$

$$\text{Average Staffing (ST)} = \text{PM} / \text{DM}$$

SLOC 11,000

Project Type: Semi-Detached

Coefficient = 3.0

T (Schedule Coefficient) = 0.35

Complexity Exponent(P): 1.12

Effort, PM = $3.0 \times (11)^{1.12} = 44$ person-months

Development Time, DM = $2.50 \times (44)^{0.35} = 9.4$ Months

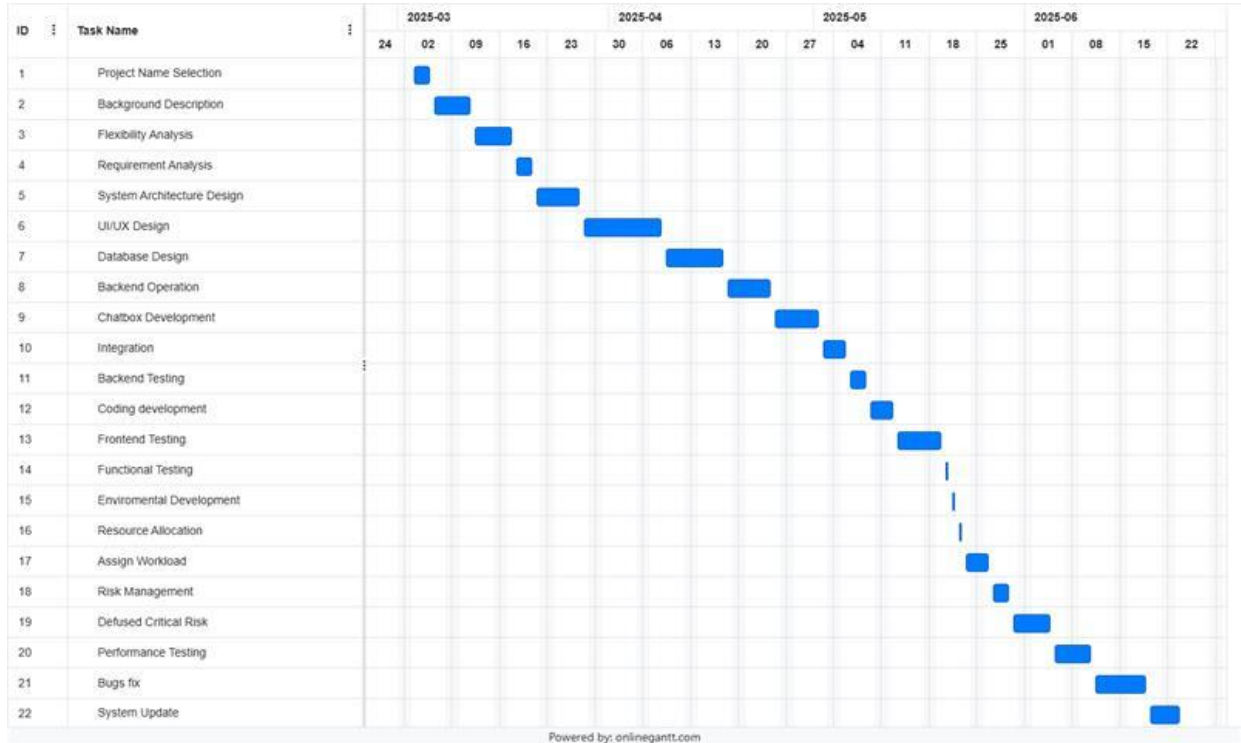
Average Staffing, ST = PM/DM

$$= 44/9.4$$

$$= 4.68 \text{ people} \sim 5 \text{ people}$$

3.3 Project Gantt Chart

Our project, Emergency Service System – Rapid Aid, is scheduled to start on 2nd March 2025, and is estimated to be completed by 22nd June 2025, resulting in a total project duration of approximately 4 months. This timeline forms the basis for our effort estimation using the COCOMO Semi-Detached mode.



Section 4: Project Risk Analysis

Risk Table:

Impact Values -

- 1 – Negligible
- 2 – Minor
- 3 – Marginal
- 4 – Critical
- 5 – Catastrophic

Risk ID	Description	Category	Probability	Impact (1-5)	RMMM (Risk Mitigation, Monitoring and Management)
ID1	System overload during emergencies	TE	60%	5	Design scalable systems with proper redundancy and continuously monitor system performance during emergencies, and implement incident response plans to manage overload situations effectively
ID2	Data breach of sensitive information	BU	30%	4	Encrypt data and regularly test security measures
ID3	Inaccurate location data for services	CU	20%	3	Integrate with reliable location tracking services and monitor accuracy metrics properly
ID4	Delayed response time due to technical issues	CU	10%	4	Optimize system performance, monitor latency, and establish rapid issue response protocols.
ID5	Hardware failure	DE	20%	2	Maintain hardware redundancy, monitor system health, and maintain proactive replacement schedules.
ID6	Limited scalability for peak demand	PS	40%	4	Optimize usage algorithms and implement proper storage redundancy to monitor

					usage trends, and plan capacity upgrades proactively.
ID7	Software bugs causing downtime	TE	30%	1	Conduct frequent software testing and monitor system stability, and maintain proper patching
ID8	Insufficient training for operators	ST	50%	2	Provide regular training sessions, monitor performance gaps, and update learning materials as needed
ID9	Power outages affecting system availability	DE	70%	3	Use backup power systems, UPS or generators and monitor power status, and implement proper failover mechanisms
ID10	Communication failure	CU	40%	3	Establish redundant communication channels, monitor connectivity, and implement clear communication protocols
ID11	Limited integration with external services	TE	70%	2	Use standardized APIs, maintain proper integration procedure, and maintain regular updates for compatibility
ID12	Cyberattacks compromising system security	TE	30%	2	Implement strong cybersecurity measures, continuously monitor threats, and conduct regular vulnerability assessments
ID13	Unavailability of critical updates	CU	40%	2	Maintain good contact with software vendors and monitor update status regularly, and ensure updates are tested properly for better compatibility
ID14	Low user adoption of the system	CU	20%	1	Focus on user centric design, provide initiative training, and gather continuous feedback for improvements
ID15	Legal or ethical issues in using AI-	TE	10%	1	Ensure proper explainability in AI

	based recommendations				driven decisions, include disclaimers and maintain regular updates for compatibility
ID16	Failure to meet regulatory compliance	PR	20%	1	Conduct regular audits and maintain compliance reports, monitor new regulations and stay updated
ID17	Insufficient bandwidth for data transfer	DE	20%	2	Upgrade network structure, monitor bandwidth usage, and prioritize critical data traffic
ID18	Incorrect prioritization of incidents	PR	30%	3	Implement clear prioritization criteria and proper incident classification algorithm train staff, and continuously improve response prioritization logic
ID19	High operational costs	BU	40%	4	Optimize resource use, monitor expenses regularly, and implement cost-effective ways
ID20	Public distrust in the system	CU	10%	4	Launch public trust building campaigns, ensure proper transparency and engage in community feedback for trust improvement
ID21	Misuse of the emergency system	PR	80%	4	Implement authentication and monitoring mechanism, monitor system access continuously
ID22	Language or accessibility barriers for diverse users	CU	30%	3	Implement multilingual support, voice commands, screen readers, and ensure accessibility compliance just like WCAG
ID23	Failure of AI model to adapt to evolving emergency trends	TE	30%	4	Periodically retrain models with new datasets, apply continuous monitoring of AI performance, and maintain human override.

ID24	Low digital literacy among target users	CU	40%	3	Design user-friendly interfaces with simple language, offer in-app tutorials and onboarding help guides
ID25	Over-dependence on mobile platforms	BU	40%	1	Develop lightweight web-based versions or USSD fallback for users with basic phones. Ensure cross-platform compatibility.
ID26	Server downtime due to cloud related issues	DE	20%	4	Use multi-region deployment, implement load balancing and automatic failover mechanisms using reliable cloud providers.
ID27	Delays in legal approvals for public deployment	PR	20%	3	Engage legal experts early, ensure documentation complies with national digital and emergency laws.
ID28	Inconsistent or inaccurate data from third-party APIs	TE	30%	3	Validate external data sources, log inconsistencies, and switch to backup APIs when error thresholds are crossed.
ID29	Misinterpretation of alert messages by users	CU	30%	3	Use clear, localized language in alerts, avoid technical jargon, and perform user testing on alert effectiveness.

Probability – Impact Matrix:

		Impact				
		Negligible	Minor	Marginal	Critical	Catastrophic
Probability	Rare	Low	Low	Low	Medium	Medium
	Unlikely	Low	Low	Medium	Medium	Medium
	Moderate	Low	Medium	Medium	Medium	High
	Likely	Medium	Medium	Medium	High	High

	Very likely	Medium	Medium	High	High	High
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